Received: June 2, 2026. Revised: July 31, 2026.

INASS

International Journal of
Intelligent Engineering & Systems

http://www.inass.org/

GA-SAR-Det: A Multi-scale YOLOv11 Architecture for Small and Densely Distributed SAR Ship Target Detection

Yudi Eko Windarto[1]* Kuntoro Adi Nugroho[1] Patricia Evericho Mountaines[1]
Bellia Dwi Cahya Putri[1] Zeka Emo[1] M Ma'ruf Sabili Riziq[1]

[1]*Universitas Diponegoro, Indonesia*
* Corresponding author's Email: yudi@live.undip.ac.id

Abstract: Synthetic Aperture Radar (SAR) ship detection remains challenging due to speckle noise, small target size, and complex maritime backgrounds, particularly in nearshore and densely distributed environments. Existing SAR ship detectors often experience degraded performance on small targets because fine-grained spatial information is progressively lost during feature extraction and multi-scale fusion. This paper proposes GA-SAR-Det, an improved YOLOv11n-based architecture centered on a novel Fibonacci-inspired channel progression strategy for enhancing multi-scale feature representation. Rather than relying on conventional channel scaling, the proposed strategy progressively allocates feature channels according to a Fibonacci recurrence, aiming to preserve feature continuity. To fully exploit the redesigned channel allocation, the proposed architecture integrates C3Ghost modules, a four-scale bidirectional PAN-FPN for enhanced multi-scale feature fusion, and a dedicated P2 detection head for high-resolution small-object localization. In addition, the CIoU regression loss is replaced with PIoUv2 to improve bounding-box optimization stability for small SAR targets. Experiments on the SSDD and HRSID public benchmarks show that GA-SAR-Det achieves mAP@0.5 scores of 98.5% and 91.5%, respectively, outperforming the retrained YOLOv11n baseline under identical experimental settings. These results demonstrate that the proposed Fibonacci-inspired channel progression, together with the supporting architectural enhancements, effectively improves the detection of small and densely distributed ship targets in complex SAR maritime environments.

Keywords: SAR ship detection, Small object detection, YOLOv11, C3Ghost, PAN-FPN, P2 detection head, PIoUv2, SSDD, HRSID.

1. Introduction

Synthetic Aperture Radar (SAR) has emerged as an indispensable sensor modality for maritime domain awareness, owing to its all-weather, day-and-night active microwave imaging capability that penetrates cloud cover, atmospheric haze, and adverse sea states [1]. Unlike passive optical sensors, SAR generates high-resolution reflectivity maps of ocean surfaces by exploiting advanced synthetic aperture processing of coherently received radar echoes, enabling continuous and reliable monitoring of vessel traffic regardless of illumination or meteorological conditions [2]. Consequently, SAR imagery has become the primary data source for a broad range of maritime applications, including illegal fishing detection, oil-spill monitoring, and national-security surveillance.

Despite these operational advantages, automated ship detection in SAR images remains a fundamentally challenging problem driven by three interrelated physical phenomena. First, SAR is an inherently coherent imaging system whose received signal is formed by the superposition of echoes from multiple elementary scatterers, producing a multiplicative granular noise-speckle-that degrades image quality and significantly reduces feature saliency by masking target brightness relative to background clutter [3]. Second, ship targets in typical surveillance scenarios occupy only a few tens of pixels after successive backbone down-sampling, making precise localization difficult and increasing the risk of missed detections [4]. Third, near-shore

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

and port environments introduce dense man-made infrastructure whose radar cross-section rivals that of actual vessels, dramatically elevating false-alarm rates.

General-purpose detectors-Faster R-CNN, RetinaNet, Cascade R-CNN, and the YOLO series-achieve impressive results on natural-image benchmarks such as COCO and PASCAL VOC, yet their direct application to SAR imagery yields sub-optimal performance [5-7]. Two-stage methods (e.g., Faster R-CNN, Cascade R-CNN) offer high localization accuracy but incur prohibitive inference latency that precludes real-time on-board deployment. Single-stage models (e.g., YOLOv8, RetinaNet) achieve higher throughput but suffer from degraded sensitivity to small, low-contrast targets whose features are overwhelmed by SAR speckle characteristics [6, 7]. Furthermore, high-accuracy architectures typically carry large parameter budgets that exceed the memory constraints of unmanned aerial vehicle (UAV) payloads and satellite ground stations operating under resource-constrained conditions.

Recent literature has explored complementary strategies to partially address these limitations: (i) convolutional primitives such as ghost convolutions and depth-wise separable convolutions that reduce computational overhead [8-12]; (ii) channel and spatial attention mechanisms that re-weight feature responses to suppress sea-clutter interference[13-19]; and (iii) frequency-domain feature extraction via wavelet transforms that preserve high-frequency edge detail lost by conventional pooling[20, 21]. Despite these advances, detecting small ships in SAR imagery remains challenging due to the limited spatial representation of tiny targets and the loss of fine-grained features during feature extraction and fusion. Furthermore, many existing SAR detectors prioritize parameter reduction and inference speed, which may limit their ability to preserve fine-grained features required for detecting small and densely distributed ship targets.

To bridge this gap, this paper proposes GA-SAR-Det, a multi-scale grouped-attention SAR ship detector built upon a redesigned YOLOv11n architecture. The principal contributions of this work are:

(1) A Fibonacci-inspired channel progression strategy is introduced as an alternative channel allocation mechanism for YOLOv11, and is integrated with C3Ghost modules to improve multi-scale feature representation.
(2) A four-scale bidirectional PAN-FPN neck architecture (P2-P5) that enhances feature aggregation between shallow spatial information and deep semantic information for improved detection of small and densely distributed ships.
(3) A dedicated high-resolution P2 detection head operating at stride-4 resolution to preserve fine-grained ship features and improve localization sensitivity for sub-32-pixel ship targets in SAR imagery.
(4) Replacement of the CIoU regression loss with PIoUv2 to improve bounding-box regression stability and localization performance for small SAR ship targets.

The remainder of this paper is organized as follows. Section 2 reviews related work on SAR ship detection, architectures, and multi-scale feature fusion. Section 3 describes the proposed GA-SAR-Det architecture, datasets, experimental settings, and evaluation metrics. Section 4 presents the ablation studies and comparative experimental results. Section 5 discusses the findings, limitations, and future research directions.

2. Related work

2.1 Traditional and deep learning approaches to SAR ship detection

Early SAR ship detection relied on handcrafted feature engineering and statistical modeling of background clutter. The Constant False Alarm Rate (CFAR) detector [22] remains the most widely studied classical algorithm; it estimates local background statistics and sets an adaptive decision threshold to separate ship returns from sea clutter. Despite its real-time computational profile, CFAR degrades significantly in textured near-shore environments and densely packed vessel configurations, where the local homogeneity assumption is violated, leading to elevated false-alarm rates and missed detections [4].

The advent of large annotated SAR datasets-most notably SSDD [23] and HRSID [24]-catalyzed the adoption of deep convolutional neural networks (CNNs) for ship detection, displacing handcrafted pipelines as the de facto standard. CNN-based detectors are broadly categorized into two-stage and single-stage frameworks, each presenting a distinct accuracy-efficiency trade-off that is particularly consequential for maritime surveillance under resource-constrained deployment conditions.

2.1.1. Two-stage detectors

Region-proposal-based methods such as R-CNN [25] and Faster R-CNN [26] achieve high localization accuracy through a decoupled detect-then-classify

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

pipeline. Domain-specific adaptations have yielded meaningful gains: C-SE Mask R-CNN [27] integrates a multi-scale context compression excitation module to extract contextual features across background scales, improving instance segmentation in cluttered port environments. Meta-RCNN [28] constructs a Class-Domain Prototype Extraction (CDPE) module to address cross-domain generalization and few-shot learning scenarios. MFT-Reasoning RCNN [29] embeds knowledge graph inference into the Faster R-CNN framework, explicitly modeling relational context between ship targets and their surroundings. The Multi-Feature Fusion Network (MFFN) [30] further augments contextual modeling through joint spatial-frequency feature fusion via a dual-branch architecture.

2.1.2. Single-stage detectors

The YOLO series [31] and SSD achieve detection in a single forward pass, offering substantially higher throughput with moderate accuracy trade-offs. SAR-specific single-stage adaptations have targeted three core weaknesses: background clutter sensitivity, small-target localization, and computational overhead. YOLO-Lite [32] introduces a Lightweight Feature Enhancement Block (LFEBNet) and Coordinate Perception Attention (CPEA) to reduce model complexity while improving sensitivity to SAR-specific textures. ST-YOLOA [33, 34] embeds a Swin Transformer module for global feature representation and a Coordinate Attention mechanism for precise target localization in cluttered environments, reporting a 4.83% improvement in detection accuracy over YOLOX on nearshore and multi-scale scenarios. ELLK-Net [34] leverages adaptive large-kernel convolutions and a Large Kernel Multiscale Attention (LKMA) module to expand the effective receptive field without a proportional parameter increase, demonstrating that receptive field expansion can be achieved efficiently. While single-stage detectors achieve favorable inference speeds, they systematically underperform two-stage counterparts on sub-32-pixel ship instances whose features are overwhelmed by SAR speckle.

2.2 Small-object-oriented architectures

Preserving discriminative features while maintaining effective feature representation is an important challenge in SAR ship detection, particularly for small targets whose visual characteristics can be easily lost during deep feature extraction. Recent studies have therefore explored alternative feature extraction modules that improve feature diversity and representation quality while reducing redundant feature generation. Among these approaches, Ghost-based architectures generate complementary feature maps from intrinsic feature representations, enabling richer multi-scale features that may benefit small-object detection.

Depthwise separable convolutions decompose a k×k operation into a per-channel depth-wise convolution followed by a 1×1 point-wise projection, reducing parameters by a factor of approximately k^2 at marginal accuracy cost [11]. Several recent SAR-specific models exploit this factorization: LHSDNet [9] applies depthwise separable operations within a multi-scale feature extraction backbone to achieve a 3× parameter reduction while maintaining competitive mAP on the SSDD benchmark. L-Enet [10] further pushes the compression boundary by combining depthwise separable convolutions with a custom detection head for ultra-compact deployment.

Ghost convolutions [12] identify and exploit feature-map redundancy more aggressively by generating a large set of "ghost" feature maps from a small set of intrinsic maps through cheap linear transforms (typically depth-wise convolutions). The C3Ghost module [21] embeds Ghost convolutions within a Cross Stage Partial (CSP) residual structure, providing a drop-in replacement for the standard C3 block in YOLOv5/v8/v11 backbones. Empirical evaluation on remote-sensing imagery [21] shows that C3Ghost reduces parameters by 40-60% with less than 0.5% mAP degradation, making it particularly suitable for preserving feature diversity while reducing feature redundancy.

Complementing these convolution-level reductions, Fibonacci-inspired channel progression strategies have been explored in aerial-image detectors to promote feature reuse across depth [35]. Rather than doubling channels at each stage in a fixed geometric sequence, Fibonacci scheduling (e.g., 32→64→96→128→256→512) connects each block to its two predecessors, analogous to DenseNet connectivity but with lower memory overhead. This design broadens the effective receptive field and improves multi-scale representation without increasing depth, a property particularly valuable when detecting sub-pixel ship instances after successive down-sampling.

Despite these advances, existing architectures treat backbone compression and attention-based feature enhancement as independent design choices. No prior work has demonstrated a unified framework that simultaneously applies Fibonacci channel progression and Ghost convolution modules, within a single SAR-optimized backbone, which motivates the design proposed in Section III.

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

2.3 Multi-scale feature fusion and frequency-domain enhancement

Multi-scale feature fusion is essential for SAR ship detection because vessels of interest span a wide range of sizes-from tankers occupying hundreds of pixels to fishing boats represented by fewer than 10 pixels-within a single image. The Feature Pyramid Network (FPN) [36] established the canonical top-down fusion paradigm, propagating high-level semantic features from deep layers to high-resolution shallow maps via lateral connections. FPN has been consistently shown to improve small-target detection in SAR imagery by ensuring that every prediction scale has access to both semantic context and spatial detail [37, 38].

The Path Aggregation Network [39, 40] augments FPN with an additional bottom-up pathway, creating bidirectional feature flow that strengthens the propagation of low-level structural features (edges, textures) back toward the deeper semantic levels. The resulting two-way connectivity substantially improves feature richness at all scales. Scale Sequence FPN (ssFPN) [39, 40] further refines this architecture by introducing sequential scale aggregation, reducing feature aliasing between adjacent pyramid levels.

A fundamental limitation of standard multi-scale pyramids is that successive max-pooling and stride-2 convolutions irreversibly discard high-frequency spatial information [20]. For sub-32-pixel ships, this information loss is catastrophic: the bounding-box boundaries and contrast edges that define a vessel's extent are precisely the high-frequency components that pooling eliminates. Adding a dedicated P2 detection head (stride 4, 128×128 output for a 640×640 input) partially mitigates this problem by branching the detection stream before excessive down-sampling occurs. Studies on aerial imagery report AP gains of up to 7.4 percentage points for small objects when a P2 head is added to a three-head YOLO baseline [41].

2.4 Summary and research gap

Table 1 summarizes representative state-of-the-art methods along the four dimensions reviewed above: detection paradigm and multi-scale design. Several observations emerge from this analysis. First, existing detectors often struggle to simultaneously preserve high-resolution spatial information and strong semantic representations for small ship targets. Second, existing recent single-stage detectors (SOD-YOLO, LSD-Det) achieve competitive mAP@0.5 but exhibit limited Recall on HRSID, suggesting insufficient small-object sensitivity after domain transfer. Third, while previous studies have primarily improved YOLO-based SAR ship detectors through lightweight backbone modules, feature-fusion networks, attention mechanisms, or additional detection heads, the channel progression strategy itself has remained largely unexplored as a design dimension. In particular, existing methods continue to adopt conventional channel scaling schemes, and no prior work has investigated Fibonacci-inspired channel progression for feature extraction in YOLO-based SAR ship detection.

Motivated by this gap, the proposed GA-SAR-Det is designed with the following key components:

(a) A novel Fibonacci-inspired channel progression strategy for channel allocation, enabling smoother feature-capacity scaling across backbone stages;
(b) A C3Ghost-based backbone that integrates the proposed channel progression to improve feature extraction;
(c) A bidirectional PAN-FPN neck to strengthen multi-scale feature fusion between shallow and deep representations; and
(d) A dedicated P2 high-resolution detection head to enhance the detection of small ship targets.

3. Materials and methods

3.1 Overall architecture of GA-SAR-Det

The proposed GA-SAR-Det is derived from YOLOv11n and is designed to improve ship detection in Synthetic Aperture Radar (SAR) imagery, where small targets, speckle noise, and cluttered backgrounds present significant challenges.

Table 1. Representative SAR Ship Detection Methods

Method	Type	Neck	P2 Head	mAP @0.5 (SSDD)
Faster R-CNN [13]	Two-stage	FPN	No	~94.0
YOLO-Lite [19]	Single-stage	MFFNet (Multi-scale Feature Fusion Network)	No	98.4
SOD-YOLO [10]	Single-stage	BSSI-FPN (extended PAN-FPN)	No	97.2
GA-SAR-Det (Ours)	Single-stage	Bidirectional PAN-FPN	Yes	98.5

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

As illustrated in Fig. 1, the proposed architecture is centered on a novel Fibonacci-inspired channel progression strategy, which redesigns the conventional channel allocation of the YOLOv11 backbone. Unlike existing YOLO-based SAR ship detectors that primarily improve feature extraction through lightweight modules or enhanced feature-fusion networks, the proposed approach introduces channel progression itself as a new architectural design dimension. By adopting a Fibonacci-inspired channel scaling scheme, the backbone aims to achieve smoother feature-capacity transitions across network stages, promoting improved feature continuity and multi-scale representation.

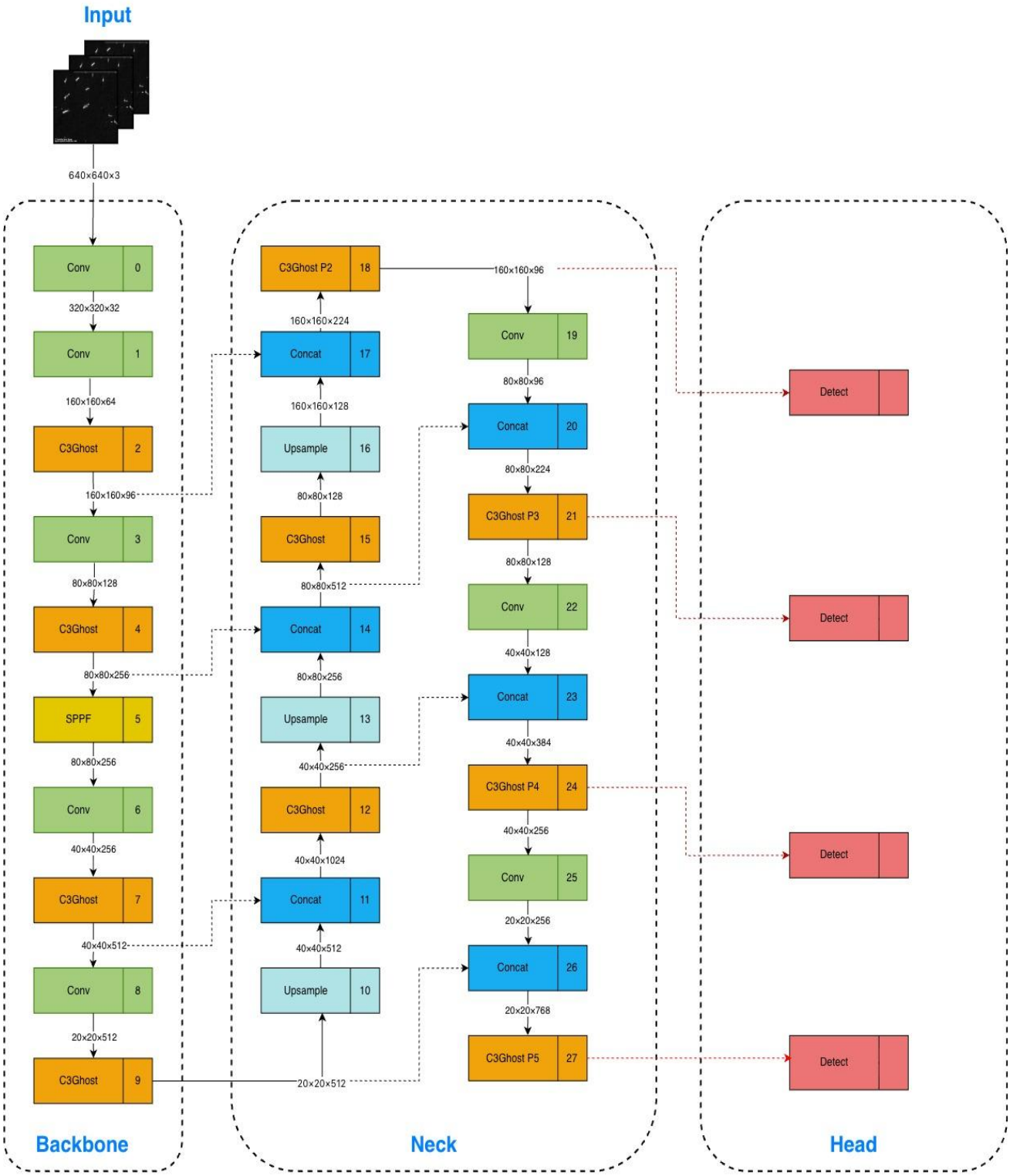


Figure. 1 GA-SAR-Det Architecture

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

To effectively exploit the proposed channel progression, the backbone integrates C3Ghost modules to improve feature reuse with minimal computational overhead. Furthermore, the neck is redesigned as a four-scale bidirectional PAN-FPN (P2-P5), enabling richer feature interaction across semantic levels. Finally, a dedicated P2 detection head is introduced to preserve high-resolution feature maps for detecting tiny ships. Together, these complementary architectural components enhance fine-grained spatial representation while allowing the proposed Fibonacci-inspired channel progression to be effectively utilized throughout the detection pipeline. Table 2 summarizes the structural differences between GA-SAR-Det and the YOLOv11n baseline.

3.1.1. Input pre-processing layer

The input layer of the improved YOLOv11n model processes the raw image $X \in R^{c\times h\times w}$ and applies channel-wise normalization followed by bilinear resizing to the target resolution of 640×640 pixels. This stage is identical to the YOLOv11n baseline and introduces no learnable parameters. Standard mosaic augmentation, random horizontal flip, and HSV colour-space jitter are applied during training to improve domain generalization across SSDD and HRSID imaging conditions.

3.1.2. Backbone: Fibonacci channel progression with C3Ghost

The backbone is responsible for hierarchical feature extraction and forms the foundation of the proposed detector. Similar to YOLOv11, it employs standard Conv-BN-SiLU blocks, an initial stem convolution for spatial downsampling, and an SPPF module to enlarge the receptive field. The principal modification introduced in this work is a novel Fibonacci-inspired channel progression strategy, which replaces the conventional channel scaling used by the baseline architecture. Rather than relying on standard channel progression, the proposed design allocates feature channels according to a Fibonacci-inspired sequence (e.g., 32 → 64 → 96 → 128 → 256 → 512), producing smoother transitions in representational capacity across successive backbone stages. This design is intended to encourage more effective feature reuse and improve the preservation of multi-scale information, particularly for small ship targets in SAR imagery.

To implement the proposed channel progression, the conventional C3 blocks are replaced with C3Ghost modules. The Ghost operations reduce redundant feature generation while complementing the proposed channel allocation strategy, enabling richer feature representation without substantially increasing computational complexity.

3.1.3. Neck: four-scale PAN-FPN

The neck combines features extracted by the backbone at different scales. As in YOLOv11, the model incorporates the PAN-FPN (Path Aggregation + Feature Pyramid Network) to achieve bidirectional fusion of low- and high-level features. To complement the proposed backbone design, the neck introduces several architectural modifications. While the baseline YOLOv11 utilizes a three-scale PANet (P3-P5), our proposed neck extends this to a four-scale P2-P5 bidirectional architecture, ensuring that high-resolution spatial details are preserved and aggregated across all semantic levels. This extension results in a four-scale detection framework (P2-P5) that enhances the detection of tiny objects such as ships in SAR images. On the bottom-up pathway, the model introduces an extra refinement path (P2 → P5) to strengthen feature aggregation and employs a fusion mechanism that combines standard convolutional outputs with C3Ghost outputs, prioritizing feature richness for small-target sensitivity, at the cost of additional parameters compared with the baseline neck.

The detection head is responsible for predicting bounding boxes, objectness scores, and class probabilities. Consistent with YOLOv11, it adopts a decoupled head, where classification and regression are handled separately, and applies loss functions based on IoU and cross-entropy for optimization. This stage operates on four detection scales (P2-P5), instead of three detection scales (P3-P5) used in standard YOLOv11.

Table 2. Architectural Comparison between YOLOv11n and GA-SAR-Det

Component	YOLOv11n (Baseline)	GA-SAR-Det (Proposed)
Backbone Block	C3 (standard CSP)	C3Ghost + Fibonacci channel progression
Channel Schedule	32→64→128→256→512	32→64→96→128→256→512
Neck Module	Standard C2f bottleneck	Bidirectional PAN-FPN neck
Detection Scales	3-scale: P3, P4, P5	4-scale: P2, P3, P4, P5
Regression Loss	CIoU	PIoUv2

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

This extension significantly enhances the model's ability to detect very small targets in SAR images, while maintaining accuracy for larger objects.

3.2 C3Ghost module

Standard convolutional layers with C_1 input channels and C_2 output channels require $P = k^2 \cdot C_1 \cdot C_2$ parameters, which becomes prohibitively large for deep, wide networks operating on resource-constrained edge hardware. The C3Ghost module [21] addresses this by restructuring feature generation into two complementary steps: intrinsic feature extraction via a compact standard convolution, and ghost feature synthesis via inexpensive depth-wise linear transforms. Together, these steps produce an output with the same channel count as a full convolution at substantially reduced parameter and FLOPs cost.

3.2.1. Intrinsic feature extraction

Given an input tensor $X \in R^{c \times h \times w}$, a standard convolution with filter $f \in R^{c \times k \times k \times n}$ generates $m = C_2/2$ intrinsic feature maps:

$$Y' = \sigma\big(BN(X * f + b)\big) \in \mathbb{R}^{\{H' \times W' \times m\}} \quad (1)$$

where $(*)$ denotes convolution, b is the bias vector, BN is batch normalization, and σ is the SiLU activation function. The FLOPs cost of this step is $\Phi_1 = m \cdot H' \cdot W' \cdot C \cdot k^2$, which is halved relative to a full C_2-channel convolution by setting $m = C_2/2$.

3.2.2. Ghost feature synthesis

Each of the m intrinsic maps $y'_i \in \mathbb{R}^\{H' \times W'\}$ is transformed by $s - 1 = 1$ depth-wise convolutions $\Phi_{i,j}$ to generate ghost feature maps:

$$y_{i,j} = \Phi_{i,j}(y'_i), \forall i = 1, \dots, m, j = 1, \dots, s, \quad (2)$$

The final output $Y \in \mathbb{R}^\{H' \times W' \times C_2\}$ is obtained by concatenating the intrinsic maps $\{y'_i\}$ and all ghost maps $\{y_{i,j}\}$ along the channel dimension:

$$Y = Concat([Y', \{y_{i,j}\}] \in \mathbb{R}^\{H' \times W' \times C_2\} \quad (3)$$

3.3 Fibonacci-inspired channel progression

While C3Ghost reduces redundant computation at the module level, the cumulative effect of the proposed architectural extensions (four-scale neck, P2 head) on overall model size is addressed separately in Section 5. To improve multi-scale feature representation across the network, GA-SAR-Det introduces a novel Fibonacci-inspired channel progression strategy throughout both the backbone and neck architecture. Unlike existing YOLO-based improvements that primarily modify convolutional modules, feature-fusion networks, or detection heads, the proposed method redesigns the channel allocation strategy itself, introducing channel progression as an additional architectural design dimension. The Fibonacci recurrence was selected because it naturally produces gradually increasing channel widths while avoiding abrupt capacity jumps associated with exponential channel scaling. This provides a simple deterministic rule that improve representational growth. Accordingly, the channel width at the $k-$ th stage is defined as

$$C_k = C_{k-1} + C_{k-2} \quad (4)$$

where C_k denotes the channel width at the $k-$th stage. Following Eq. (4), the proposed architecture adopts the channel progression

$$32 \rightarrow 64 \rightarrow 96 \rightarrow 128 \rightarrow 256 \rightarrow 512 \quad (5)$$

Compared with conventional exponential channel scaling, the proposed Fibonacci-inspired progression provides a more gradual increase in representational capacity across successive stages. This smoother allocation is intended to preserve intermediate feature representations while progressively enriching semantic information, improving the balance between spatial detail and high-level feature abstraction. Consequently, feature maps from both shallow and deep layers can contribute more effectively during multi-scale feature aggregation, which is particularly beneficial for detecting small and densely distributed ship targets in SAR imagery.

The proposed channel progression is consistently applied throughout both the backbone and neck of GA-SAR-Det. C3Ghost modules are employed to efficiently implement the proposed channel allocation, whereas the redesigned bidirectional PAN-FPN and the additional P2 detection head are incorporated to fully exploit the enriched multi-scale representations produced by the backbone. Unlike conventional architectural modifications that replace individual network components, the proposed channel progression redistributes representational capacity across the entire feature hierarchy.

As channel allocation governs the information-processing capability of every network stage, this design influences feature extraction throughout the

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

backbone rather than affecting only a single module. Accordingly, the remaining architectural components are intended to complement and maximize the effectiveness of the proposed Fibonacci-inspired channel progression.

3.4 PIoUv2 loss

Bounding-box regression quality significantly affects localization performance in SAR ship detection, particularly for small targets whose boundaries are often blurred by speckle noise and low contrast. The original YOLOv11n model employs Complete-IoU (CIoU) Loss for bounding-box optimization. However, CIoU relies heavily on center-point distance and aspect-ratio consistency, which may become unstable for small SAR targets during early training stages.

To improve localization stability, this study adopts the PIoUv2 loss function proposed in [42] as a replacement for CIoU in the regression branch. Unlike CIoU, PIoUv2 evaluates localization quality through direct boundary alignment by measuring discrepancies between the predicted and ground-truth box boundaries. The regression loss is built upon the standard IoU loss,

$$IoU = \frac{|B \cap B_{gt}|}{|B \cup B_{gt}|} \tag{6}$$

$$L_{IoU} = 1 - IoU \tag{7}$$

In Eq. (6), B and B_{gt}represent the predicted and ground-truth bounding boxes, respectively. To further improve localization accuracy, the IoU loss is augmented with a boundary-aware penalty term:

$$L_{\{PIoU\}} = L_{IoU} + 1 - e^{\{-\rho^2\}} \tag{8}$$

where ρ denotes the normalized boundary deviation between the predicted and ground-truth boxes. By focusing on boundary consistency rather than center-point distance, PIoU is more robust to contour ambiguity commonly observed in SAR imagery.

PIoUv2 further introduces an anchor-quality attention mechanism that dynamically adjusts the contribution of each training sample. The final loss is expressed as

$$L_{PIoUv2} = u(\lambda q), L_{PIoU} \tag{9}$$

where q is the anchor quality score, λ) is a scaling factor, and $\big(u(\cdot)\big)$ is a non-monotonic weighting function. This mechanism emphasizes moderately informative anchors while suppressing overly easy or highly misaligned samples, resulting in faster convergence and more stable optimization. Consequently, PIoUv2 provides improved localization performance for SAR ship detection, particularly in scenarios involving small targets and blurred object boundaries.

3.5 Dataset

Two public datasets for SAR ship detection were used in this study:

(1) SAR Ship Detection Dataset (SSDD) [23] consists of 1,160 SAR images sourced from Sentinel-1, TerraSAR-X, and RadarSat-2 satellites. For this study, the SSDD dataset is divided into an 8:2 ratio for training and testing, respectively.

(2) High-Resolution SAR Images Dataset for Ship Detection (HRSID) [24] is compiled from Sentinel-1 and TerraSAR-X imagery. The dataset consists of 5,604 cropped SAR image patches derived from 136 large-scale panoramic scenes**.** For this study, the 5,604 HRSID image patches are partitioned into training and test sets with an 8:2 ratio, consistent with the split protocol applied to SSDD.

3.6 Experimental setup

The experiments were conducted on an Ubuntu 20.04 environment using Python 3.8, PyTorch1.11.0, and CUDA 11.3. The hardware platform consisted of a 12-vCPU Intel Xeon Platinum 8255C processor and an NVIDIA RTX 2080 Ti GPU with 11 GB of memory. All models were trained for 150 epochs with an input resolution of 640 × 640 pixels. The SSDD dataset was trained using a batch size of 8 with 2 data-loading workers, whereas the HRSID dataset used a batch size of 16 with 4 workers.

To ensure reproducibility and fair comparison, YOLOv8n, YOLOv11n, and SOD-YOLO [7] were retrained under the same experimental conditions as GA-SAR-Det. No results for these models were copied from prior publications. In contrast, LSD-Det [19], DBW-YOLO [43], YOLO-MSD [44], and ELLK-Net [34], were taken directly from their respective publications because official source codes or implementation details were not publicly available, preventing retraining under a unified protocol.

The unified training protocol consisted of the SGD optimizer with an initial learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005, together with a cosine learning-rate scheduler employing linear warmup during the first three epochs.

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

A fixed random seed of 42 was used throughout all retrained experiments. Data augmentation was applied consistently across all retrained models using mosaic augmentation (p = 1.0), random horizontal flipping (p = 0.5), and HSV color-space jitter (hue = 0.015, saturation = 0.7, value = 0.4). Both SSDD and HRSID followed an 80:20 train-test split, and the corresponding split files were fixed and shared across all retrained experiments.

To facilitate reproducibility, the source code, model configuration, trained weights, and dataset split files are available in this link: https://github.com/marufsabili/GA-SAR-Det

3.7 Evaluation metrics

Model performance is evaluated using a comprehensive set of complementary metrics that jointly characterize detection accuracy, localization precision, and small-object sensitivity.

3.7.1. Detection accuracy metrics

Precision and Recall score quantify the trade-off between false alarms and missed detections at a fixed IoU threshold of 0.5:

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \tag{10}$$

$$Recall = \frac{TP}{TP+FN} \tag{11}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively. In maritime surveillance, Recall is operationally more critical than Precision because undetected vessels (FN) pose higher safety and security risk than spurious detections (FP).

3.7.2. Mean average precision

Mean Average Precision (mAP) is the primary accuracy metric. Average Precision (AP) is the area under the Precision-Recall curve for a single object category:

$$AP = \int_0^1 P(R)\ dR \tag{12}$$

$$mAP = \frac{\sum_{i=1}^{m} AP_i}{m} \tag{13}$$

Two mAP variants are reported. mAP@0.5 evaluates detection at IoU ≥ 0.5, reflecting coarse localisation performance. mAP@0.5:0.95 averages AP over ten IoU thresholds from 0.5 to 0.95 in steps of 0.05, providing a stringent, scale-agnostic measure of localization accuracy. The latter is particularly diagnostic in SAR scenarios where bounding-box precision varies significantly with target size and speckle density.

4. Result

4.1 Ablation study

The ablation results in Table 3 provide a controlled comparison of individual architectural modifications relative to the baseline model. Among the evaluated components, C3Ghost achieves the highest Precision, Recall, and mAP@0.5 on SSDD, while also obtaining the highest Precision and mAP@0.5 among the single-component variants on HRSID. The P2 head achieves the highest Recall on HRSID (82.5%) and the second-highest mAP@0.5 (90.0%). These results indicate that C3Ghost and the P2 head consistently rank among the strongest-performing individual modifications across both datasets. In contrast, the remaining components produce smaller changes when evaluated independently, suggesting that their contribution may be more apparent when combined with other architectural elements.

Table 3. Ablation Study of Individual Architectural Components

Model	Prec		Rec		mAP@0.5	
	SSDD	HRSID	SSDD	HRSID	SSDD	HRSID
Baseline	94.4	90.7	92.9	78.2	97.2	88.7
+C3 Ghost	**96.27**	**91.60**	**96.2**	80.7	**98.29**	89.4
+P2 Head	94.1	90.3	94.8	**82.5**	97.6	**90.0**
+Fibonacci	94.6	90.9	93.4	79.0	97.4	89.0
+Bidirectional Neck	94.5	90.8	93.8	80.0	97.6	89.3
+PIoUv2	94.7	91.0	93.1	78.6	97.5	89.0

Note: Bold values indicate the best result.

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

Building upon these observations, Table 4 evaluates the cumulative effect of progressively integrating the proposed components. While the C3Ghost and P2-head configuration achieves the highest Precision and Recall on SSDD, it does not achieve the highest mAP@0.5. As additional components are incorporated, mAP@0.5 increases gradually from 90.10% to 91.50% on HRSID and from 98.40% to 98.50% on SSDD. Although the improvement is more modest on SSDD, the complete GA-SAR-Det configuration ultimately achieves the highest mAP@0.5 on both datasets, together with the highest Precision (91.48%) and Recall (85.50%) on HRSID.

The performance gains are more pronounced on HRSID than on SSDD. From the initial C3Ghost and P2-head configuration to the complete architecture, Recall increases from 82.1% to 85.5% and mAP@0.5 increases from 90.1% to 91.5% on HRSID, whereas the corresponding changes on SSDD are comparatively smaller. This trend suggests that the combined architecture may provide greater benefit on the more challenging dataset.

4.2 Comparison with state-of-the-art models

Table 5 first compares GA-SAR-Det with representative YOLO-based detectors, including YOLOv8n, YOLOv9-t, YOLOv10n, YOLOv11n, YOLOv12n, and SOD-YOLO, all retrained under the same experimental protocol. Under these controlled conditions, GA-SAR-Det achieves the highest Recall among the retrained models on both SSDD (95.6%) and HRSID (85.5%), as well as the highest mAP@0.5 (98.5%) and (91.5%) on SSDD and HRISID respectively. Also highest mAP@0.5:0.95 (65.8%) on HRSID. On HRSID, our model achieves second highest precision (91.5%) slightly behind YOLOv8n (91.6%). These results indicate that, under this controlled retraining protocol, the proposed architectural modifications improve detection sensitivity and localization performance relative to comparable YOLO baselines, particularly on the more challenging HRSID dataset.

For broader context, Table 5 also includes several recent SAR ship detection methods reported in the literature.

Table 4. Cumulative Ablation Study Result on Both Datasets

C3 Ghost	P2 Head	Fibo-nacci	Bidire-ctional Neck	PIoUv2	Precision		Recall		mAP@0.5	
					SSDD	HRSID	SSDD	HRSID	SSDD	HRSID
✓	✓	-	-	-	**95.84**	91.37	**95.7**	82.1	98.4	90.1
✓	✓	✓	-	-	95.63	91.22	95.6	83.3	98.4	90.7
✓	✓	✓	✓	-	95.3	91.35	95.6	84.0	98.45	91.0
✓	✓	✓	✓	✓	95.02	**91.48**	95.6	**85.50**	**98.5**	**91.50**

Note: Bold values indicate the best result.

Table 5. Comparison with SAR Ship Detection Models

Model	Precision		Recall		mAP@0.5		mAP@0.5:0.95	
	SSDD	HRSID	SSDD	HRSID	SSDD	HRSID	SSDD	HRSID
YOLOv8n	**96.4**	**91.6**	92.7	79.5	98.1	86.4	70.1	62.7
YOLOv9-t	95.5	89.7	90.2	79.1	97.2	89.7	**71.1**	64.0
YOLOv10n	92.4	89.7	89.0	78.8	95.8	88.3	69.2	63.1
YOLOv11n	94.4	90.7	92.9	78.2	97.2	88.7	70.9	64.3
YOLOv12n	93.5	89.7	90.1	77.3	96.4	87.4	69.2	61.0
SOD-YOLO	96.1	90.9	92.4	83.4	97.2	91.4	68.8	63.9
Ours	95.0	91.5	**95.6**	**85.5**	**98.5**	**91.5**	**71.1**	**65.8**
LSD-Det*	96.5	91.3	95.6	81.6	98.7	90.5	73.6	65.3
DBW-YOLO*	98.4	90.0	97.6	80.5	99.2	88.8	-	-
YOLO-MSD*	96.8	90.2	90.2	78.8	98.8	90.2	-	-
ELLK-Net*	90.3	89.6	92.5	87.7	89.7	79.7	-	-

Note: Methods marked with (*) are reported from their corresponding publications because their implementations were unavailable for retraining. Therefore, comparisons with these methods are intended as reference-level comparisons and may be influenced by differences in experimental settings. A dash (-) denotes that the corresponding metric was not provided in the original paper. Bold values indicate the best results.

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

LSD-Det was evaluated under a similar experimental protocol, whereas DBW-YOLO, YOLO-MSD, and ELLK-Net could not be retrained under our experimental settings because their implementations were unavailable. Consequently, comparisons with these methods should be interpreted as reference-level rather than strictly controlled benchmarks. Within this broader, reference-level comparison, GA-SAR-Det remains competitive rather than universally superior. On SSDD, GA-SAR-Det does not achieve the best reported Recall or mAP@0.5: DBW-YOLO reports a higher Recall (97.6% vs. 95.6%), Precision (98.4%), and mAP@0.5 (99.2%), while LSD-Det (98.7%) and YOLO-MSD (98.8%) also report higher mAP@0.5 than GA-SAR-Det (98.5%). On HRSID, GA-SAR-Det achieves the highest mAP@0.5 (91.5%) and mAP@0.5:0.95 (65.8%) among all compared methods, although ELLK-Net reports a higher Recall (87.7% vs. 85.5%). These results indicate that GA-SAR-Det is particularly competitive on HRSID, whereas on SSDD several literature-reported methods achieve higher scores on individual metrics, and that differences in the original experimental settings may influence cross-paper comparisons. Taken together, these results position GA-SAR-Det as competitive with, rather than uniformly superior to, state-of-the-art SAR ship detectors, with clear advantages specifically on the HRSID benchmark.

4.3 Visualization

To qualitatively evaluate detection performance, Fig. 2 compares the proposed GA-SAR-Det with the YOLOv11n baseline across four challenging SAR ship detection scenarios: dense distributions, offshore small targets, inshore environments, and overlapping ships. GA-SAR-Det consistently produces tighter bounding boxes around sub-32-pixel targets and demonstrates fewer missed detections in dense arrangements, which may be associated with the inclusion of the P2 high-resolution detection head.

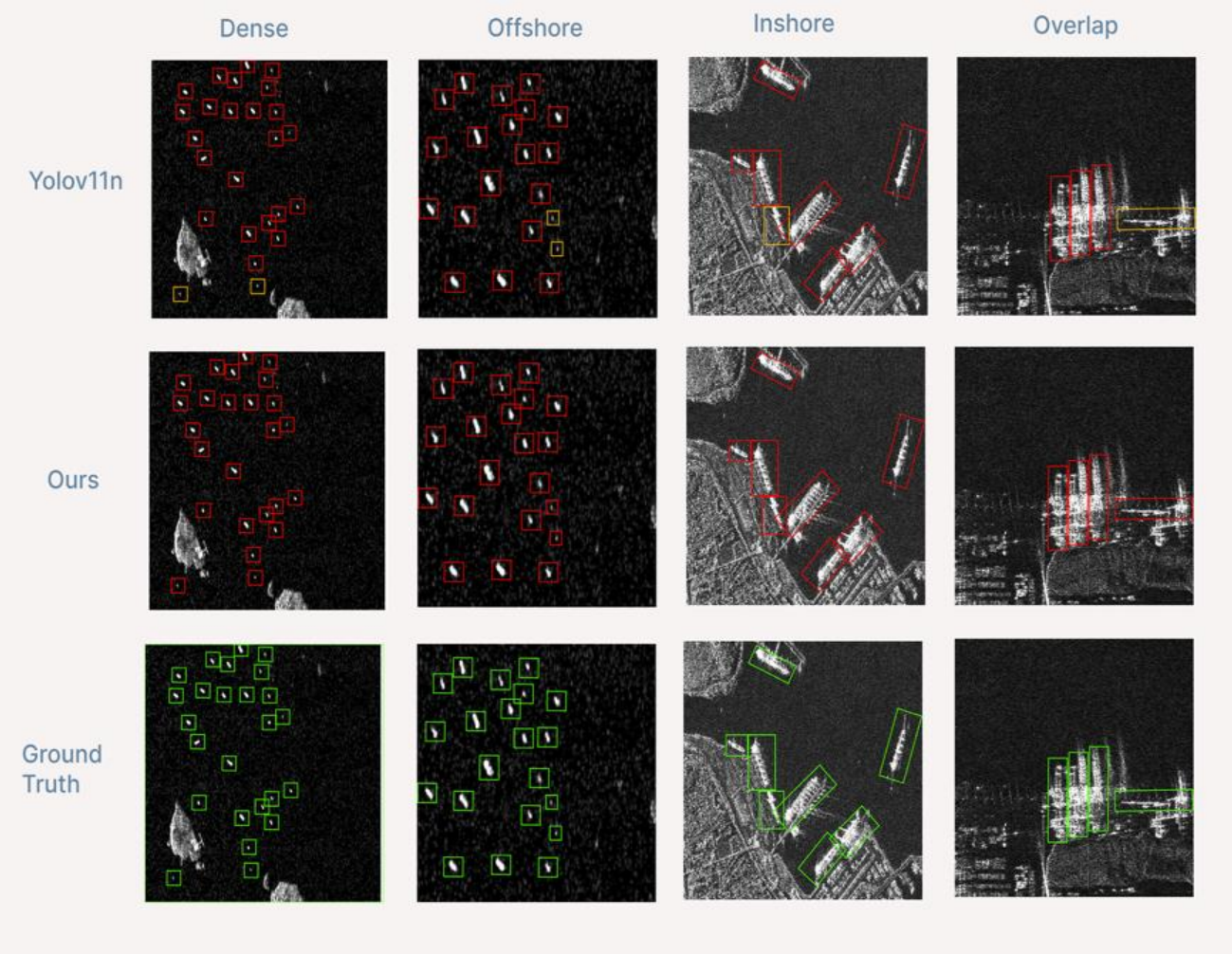


Figure. 2 Qualitative comparison between YOLOv11n and the proposed GA-SAR-Det

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

The YOLOv11n baseline exhibits systematic false negatives in nearshore scenes where multiple vessels occupy adjacent feature-map cells, consistent with the Recall improvement observed on HRSID.

5. Conclusions

The experimental results demonstrate that GA-SAR-Det achieves improved SAR ship detection performance, particularly on the more challenging HRSID dataset. Compared with the YOLOv11n baseline, the proposed model increases Recall from 78.2% to 85.5% and mAP@0.5 from 88.7% to 91.5%. The ablation study further shows that the complete architecture achieves the highest overall performance among the evaluated configurations. Together with the qualitative results, these findings indicate that the proposed design is effective for detecting ships in challenging SAR scenes and is consistent with the objective of improving sensitivity to small and densely distributed targets.

Limitations and Future Work. The proposed model increases the parameter count from 2.58M in YOLOv11n to 8.8M. Although individual components such as C3Ghost are designed to reduce redundant computation at the module level, the overall architecture is not lightweight relative to the YOLOv11n baseline, and several competing methods in Table 5 may have fewer parameters or lower computational cost; this trade-off has not been quantified in the present study. This study does not provide a comprehensive evaluation of computational efficiency, such as inference latency, memory consumption, FPS, or deployment performance on resource-constrained hardware platforms. Consequently, the trade-off between detection performance and deployment cost remains an open question. Future work will therefore focus on improving model efficiency through techniques such as pruning, knowledge distillation, and lightweight feature-fusion strategies while preserving the observed detection gains.

Conflicts of Interest

The authors declare no conflict of interest.

Author Contributions

Conceptualization, Y.E.W., K.A.N., and P.E.M.; methodology, Y.E.W., K.A.N., Z.E., and M.S.R.; software, Z.E. and M.S.R.; validation, Z.E., M.S.R., Y.E.W., and K.A.N.; formal analysis, Z.E., M.S.R., Y.E.W., and P.E.M.; investigation, Z.E., M.S.R., and B.D.C.P.; resources, Y.E.W., K.A.N., and P.E.M.; data curation, Z.E., M.S.R., and B.D.C.P.; writing-original draft preparation, Z.E., M.S.R., and Y.E.W.; writing-review and editing, Y.E.W., K.A.N., P.E.M., and B.D.C.P.; visualization, Z.E., M.S.R., and B.D.C.P.; supervision, Y.E.W., K.A.N., P.E.M., and B.D.C.P.; project administration, Y.E.W. and K.A.N.

Acknowledgments

This research was financially supported by the Faculty of Engineering, Universitas Diponegoro, Indonesia, through the Flagship Research Grant 2026.

References

[1] D. S. Ilcev, “Space Remote Sensing and Detecting Systems of Oceangoing Ships”, *Transactions on Maritime Science*, Vol. 9, No. 2, 2020, doi: 10.7225/toms.v09.n02.004.

[2] L. Liao, Q. Zhao, and W. Song, “Monitoring of Oil Spill Risk in Coastal Areas Based on Polarimetric SAR Satellite Images and Deep Learning Theory”, *Sustainability*, Vol. 15, No. 19, p. 14504, 2023, doi: 10.3390/su151914504.

[3] A. Passah, K. Amitab, and D. Kandar, “SAR Image Despeckling Using Deep CNN”, *IET Image Processing*, Vol. 15, No. 6, pp. 1285-1297, 2021, doi: 10.1049/ipr2.12104.

[4] Z. Chen, C. Liu, V. Filaretov, and D. Yukhimets, “Multi-Scale Ship Detection Algorithm Based on YOLOv7 for Complex Scene SAR Images”, *Remote Sensing*, Vol. 15, No. 8, p. 2071, 2023, doi: 10.3390/rs15082071.

[5] X. Du and X. Wu, “Small Object Detection in Synthetic Aperture Radar with Modular Feature Encoding and Vectorized Box Regression”, *Remote Sensing*, Vol. 17, No. 17, p. 3094, 2025, doi: 10.3390/rs17173094.

[6] H. Zhu, Y. Xie, H. Huang, C. Jing, Y. Rong, and C. Wang, “DB-YOLO: A Duplicate Bilateral YOLO Network for Multi-Scale Ship Detection in SAR Images”, *Sensors*, Vol. 21, No. 23, p. 8146, 2021, doi: 10.3390/s21238146.

[7] Y. Li, *et al.*, “SOD-YOLO: Small-Object-Detection Algorithm Based on Improved YOLOv8 for UAV Images”, *Remote Sensing*, Vol. 16, No. 16, p. 3057, 2024, doi: 10.3390/rs16163057.

[8] Y. Hao and Y. Zhang, “A Lightweight Convolutional Neural Network for Ship Target Detection in SAR Images”, *IEEE Transactions on Aerospace and Electronic Systems*, Vol. 60, No. 2, pp. 1882-1898, 2024, doi: 10.1109/TAES.2023.3344396.

[9] D. Dai, H. Wu, Y. Wang, and P. Ji, “LHSDNet: A Lightweight and High-Accuracy SAR Ship Object Detection Algorithm”, *Remote Sensing*,

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

Vol. 16, No. 23, p. 4527, 2024, doi: 10.3390/rs16234527.

[10] Y. Wang, M. Miao, and S. Zhu, "L-ENet: An Ultralightweight SAR Image Detection Network", *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, Vol. 17, pp. 8967-8978, 2024, doi: 10.1109/JSTARS.2024.3391852.

[11] J. Liu, M. Wu, K. Li, and Y. Liu, "A Lightweight Network Based on Dynamic Split Pointwise Convolution Strategy for Hyperspectral Remote Sensing Images Classification", *Remote Sensing*, Vol. 17, No. 5, p. 888, 2025, doi: 10.3390/rs17050888.

[12] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More Features from Cheap Operations", In: *Proc. of 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1577-1586, 2020, doi: 10.1109/CVPR42600.2020.00165.

[13] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-Excitation Networks", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. 42, No. 8, pp. 2011-2023, 2020, doi: 10.1109/TPAMI.2019.2913372.

[14] F. Gao, Y. He, J. Wang, A. Hussain, and H. Zhou, "Anchor-Free Convolutional Network with Dense Attention Feature Aggregation for Ship Detection in SAR Images", *Remote Sensing*, Vol. 12, No. 16, p. 2619, 2020, doi: 10.3390/rs12162619.

[15] X. Zhu, D. Cheng, Z. Zhang, S. Lin, and J. Dai, "An Empirical Study of Spatial Attention Mechanisms in Deep Networks", *arXiv Preprint*, arXiv:1904.05873, 2019, doi: 10.48550/arXiv.1904.05873.

[16] H. Shi, C. He, J. Li, L. Chen, and Y. Wang, "An Improved Anchor-Free SAR Ship Detection Algorithm Based on Brain-Inspired Attention Mechanism", *Frontiers in Neuroscience*, Vol. 16, 2022, doi: 10.3389/fnins.2022.1074706.

[17] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional Block Attention Module", In: *Proc. of European Conference on Computer Vision (ECCV)*, pp. 3-19, 2018, doi: 10.1007/978-3-030-01234-2_1.

[18] N. Yu, H. Ren, T. Deng, and X. Fan, "A Lightweight Radar Ship Detection Framework with Hybrid Attentions", *Remote Sensing*, Vol. 15, No. 11, p. 2743, 2023, doi: 10.3390/rs15112743.

[19] Z. Wang, B. Qin, and S. Gao, "LSD-Det: A Lightweight Detector for Small Ship Targets in SAR Images", *IEEE Access*, Vol. 13, pp. 133483-133496, 2025, doi: 10.1109/ACCESS.2025.3593021.

[20] H. Liu, X. Li, L. Wang, Y. Zhang, Z. Wang, and Q. Lu, "MS-YOLOv11: A Wavelet-Enhanced Multi-Scale Network for Small Object Detection in Remote Sensing Images", *Sensors*, Vol. 25, No. 19, p. 6008, 2025, doi: 10.3390/s25196008.

[21] R. Xiao, H. Wang, L. Wang, and H. Yuan, "C3Ghost and C3k2: Performance Study of Feature Extraction Module for Small Target Detection in YOLOv11 Remote Sensing Images", In: *Proc. of Second International Conference on Big Data, Computational Intelligence, and Applications (BDCIA 2024)*, p. 139, 2025, doi: 10.1117/12.3059792.

[22] X. Qin, S. Zhou, H. Zou, and G. Gao, "A CFAR Detection Algorithm for Generalized Gamma Distributed Background in High-Resolution SAR Images", *IEEE Geoscience and Remote Sensing Letters*, Vol. 10, No. 4, pp. 806-810, 2013, doi: 10.1109/LGRS.2012.2224317.

[23] T. Zhang, *et al.*, "SAR Ship Detection Dataset (SSDD): Official Release and Comprehensive Data Analysis", *Remote Sensing*, Vol. 13, No. 18, p. 3690, 2021, doi: 10.3390/rs13183690.

[24] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A High-Resolution SAR Images Dataset for Ship Detection and Instance Segmentation", *IEEE Access*, Vol. 8, pp. 120234-120254, 2020, doi: 10.1109/ACCESS.2020.3005861.

[25] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation", In: *Proc. of 2014 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 580-587, 2014, doi: 10.1109/CVPR.2014.81.

[26] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", *IEEE Transactions on Pattern Analysis and Machine Intelligence*, Vol. 39, No. 6, pp. 1137-1149, 2017, doi: 10.1109/TPAMI.2016.2577031.

[27] T. Zhang, X. Zhang, J. Li, and J. Shi, "Contextual Squeeze-and-Excitation Mask R-CNN for SAR Ship Instance Segmentation", In: *Proc. of 2022 IEEE Radar Conference (RadarConf22)*, pp. 1-6, 2022, doi: 10.1109/RadarConf2248738.2022.9764228.

[28] Q. Xiao and B. Sun, "Cross-Domain Few-Shot SAR Ship Object Detection with Meta-RCNN", In: *Proc. of 2024 IEEE International Conference on Signal, Information and Data*

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/

Processing (ICSIDP), pp. 1-5, 2024, doi: 10.1109/ICSIDP62679.2024.10868801.

[29] S. Zhan, M. Zhong, Y. Yang, G. Lu, and X. Zhou, "MFT-Reasoning RCNN: A Novel Multi-Stage Feature Transfer Based Reasoning RCNN for Synthetic Aperture Radar (SAR) Ship Detection", *Remote Sensing*, Vol. 17, No. 7, p. 1170, 2025, doi: 10.3390/rs17071170.

[30] S. Wang, Z. Cai, and J. Yuan, "Automatic SAR Ship Detection Based on Multifeature Fusion Network in Spatial and Frequency Domains", *IEEE Transactions on Geoscience and Remote Sensing*, Vol. 61, pp. 1-11, 2023, doi: 10.1109/TGRS.2023.3267495.

[31] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection", In: *Proc. of 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 779-788, 2016, doi: 10.1109/CVPR.2016.91.

[32] X. Ren, Y. Bai, G. Liu, and P. Zhang, "YOLO-Lite: An Efficient Lightweight Network for SAR Ship Detection", *Remote Sensing*, Vol. 15, No. 15, p. 3771, 2023, doi: 10.3390/rs15153771.

[33] K. Zhao, R. Lu, S. Wang, X. Yang, Q. Li, and J. Fan, "ST-YOLOA: A Swin-Transformer-Based YOLO Model with an Attention Mechanism for SAR Ship Detection under Complex Background", *Frontiers in Neurorobotics*, Vol. 17, 2023, doi: 10.3389/fnbot.2023.1170163.

[34] J. Shen, L. Bai, Y. Zhang, M. Chowdhuray Momi, S. Quan, and Z. Ye, "ELLK-Net: An Efficient Lightweight Large Kernel Network for SAR Ship Detection", *IEEE Transactions on Geoscience and Remote Sensing*, Vol. 62, pp. 1-14, 2024, doi: 10.1109/TGRS.2024.3451399.

[35] H. Gong, *et al.*, "Swin-Transformer-Enabled YOLOv5 with Attention Mechanism for Small Object Detection on Satellite Images", *Remote Sensing*, Vol. 14, No. 12, p. 2861, 2022, doi: 10.3390/rs14122861.

[36] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature Pyramid Networks for Object Detection", In: *Proc. of 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 936-944, 2017, doi: 10.1109/CVPR.2017.106.

[37] P. Chen, Y. Li, H. Zhou, B. Liu, and P. Liu, "Detection of Small Ship Objects Using Anchor Boxes Cluster and Feature Pyramid Network Model for SAR Imagery", *Journal of Marine Science and Engineering*, Vol. 8, No. 2, p. 112, 2020, doi: 10.3390/jmse8020112.

[38] Y. Gong, Z. Zhang, J. Wen, G. Lan, and S. Xiao, "Small Ship Detection of SAR Images Based on Optimized Feature Pyramid and Sample Augmentation", *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, Vol. 16, pp. 7385-7392, 2023, doi: 10.1109/JSTARS.2023.3302575.

[39] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path Aggregation Network for Instance Segmentation", In: *Proc. of 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 8759-8768, 2018, doi: 10.1109/CVPR.2018.00913.

[40] H.-J. Park, J.-W. Kang, and B.-G. Kim, "ssFPN: Scale Sequence (S2) Feature-Based Feature Pyramid Network for Object Detection", *Sensors*, Vol. 23, No. 9, p. 4432, 2023, doi: 10.3390/s23094432.

[41] Z. Du and Y. Liang, "Object Detection of Remote Sensing Image Based on Multi-Scale Feature Fusion and Attention Mechanism", *IEEE Access*, Vol. 12, pp. 8619-8632, 2024, doi: 10.1109/ACCESS.2024.3352601.

[42] C. Liu, K. Wang, Q. Li, F. Zhao, K. Zhao, and H. Ma, "Powerful-IoU: More Straightforward and Faster Bounding Box Regression Loss with a Nonmonotonic Focusing Mechanism", *Neural Networks*, Vol. 170, pp. 276-284, 2024, doi: 10.1016/j.neunet.2023.11.041.

[43] X. Tang, J. Zhang, Y. Xia, and H. Xiao, "DBW-YOLO: A High-Precision SAR Ship Detection Method for Complex Environments", *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, Vol. 17, pp. 7029-7039, 2024, doi: 10.1109/JSTARS.2024.3376558.

[44] S. Tian, *et al.*, "Ship Detection in Synthetic Aperture Radar Images Based on BiLevel Spatial Attention and Deep Poly Kernel Network", *Journal of Marine Science and Engineering*, Vol. 12, No. 8, p. 1379, 2024, doi: 10.3390/jmse12081379.

This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.
License details: https://creativecommons.org/licenses/by-sa/4.0/